Tidyverse and ggplot2 Analysis with Gapminder Dataset

Data Analysis with R

2025-07-07

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[[1]]
[1] "rmarkdown" "stats"     "graphics"  "grDevices" "datasets"  "utils"
[7] "methods"   "base"

[[2]]
[1] "gapminder" "rmarkdown" "stats"     "graphics"  "grDevices" "datasets"
[7] "utils"     "methods"   "base"

[[3]]
 [1] "lubridate" "forcats"   "stringr"   "dplyr"     "purrr"     "readr"
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[13] "stats"     "graphics"  "grDevices" "datasets"  "utils"     "methods"
[19] "base"

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[19] "methods"   "base"

[[5]]
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 [7] "purrr"     "readr"     "tidyr"     "tibble"    "ggplot2"   "tidyverse"
[13] "gapminder" "rmarkdown" "stats"     "graphics"  "grDevices" "datasets"
[19] "utils"     "methods"   "base"

[[6]]
 [1] "gridExtra" "openxlsx"  "readxl"    "lubridate" "forcats"   "stringr"
 [7] "dplyr"     "purrr"     "readr"     "tidyr"     "tibble"    "ggplot2"
```

```
[13] "tidyverse" "gapminder" "rmarkdown" "stats"     "graphics"  "grDevices"
[19] "datasets"  "utils"     "methods"   "base"

[[7]]
 [1] "ggExtra"   "gridExtra" "openxlsx"  "readxl"    "lubridate" "forcats"
 [7] "stringr"   "dplyr"     "purrr"     "readr"     "tidyr"     "tibble"
[13] "ggplot2"   "tidyverse" "gapminder" "rmarkdown" "stats"     "graphics"
[19] "grDevices" "datasets"  "utils"     "methods"   "base"

[[8]]
 [1] "knitr"     "ggExtra"   "gridExtra" "openxlsx"  "readxl"    "lubridate"
 [7] "forcats"   "stringr"   "dplyr"     "purrr"     "readr"     "tidyr"
[13] "tibble"    "ggplot2"   "tidyverse" "gapminder" "rmarkdown" "stats"
[19] "graphics"  "grDevices" "datasets"  "utils"     "methods"   "base"
```

Introduction

This R Markdown document demonstrates data manipulation and visualization using the `tidyverse` suite of packages, particularly `dplyr`, `tidyr`, `readr`, `stringr`, and `ggplot2`. We will use the `gapminder` dataset to perform various data operations and create visualizations. The document also includes solutions to the provided exercises.

Installing and Loading Packages

First, we load the necessary packages using renv.

```r
library(renv)
renv::restore(prompt = FALSE)
```

Exploring the Gapminder Dataset

Let's take a look at the first few rows of the `gapminder` dataset.

```r
# View first six rows
head(gapminder)
```

```
# A tibble: 6 x 6
  country     continent  year lifeExp      pop gdpPercap
  <fct>       <fct>     <int>   <dbl>    <int>     <dbl>
1 Afghanistan Asia       1952    28.8  8425333      779.
2 Afghanistan Asia       1957    30.3  9240934      821.
3 Afghanistan Asia       1962    32.0 10267083      853.
4 Afghanistan Asia       1967    34.0 11537966      836.
5 Afghanistan Asia       1972    36.1 13079460      740.
6 Afghanistan Asia       1977    38.4 14880372      786.
```

Data Manipulation with dplyr

The `dplyr` package provides functions for data manipulation, such as `select()`, `filter()`, `mutate()`, `arrange()`, `summarise()`, and `group_by()`.

Selecting Columns

Select specific columns using `select()`.

individual columns

```r
# Select individual columns
sel_data <- gapminder %>% select(country, continent, pop)
head(sel_data)
```

```
# A tibble: 6 x 3
  country     continent      pop
  <fct>       <fct>        <int>
1 Afghanistan Asia       8425333
2 Afghanistan Asia       9240934
3 Afghanistan Asia      10267083
4 Afghanistan Asia      11537966
5 Afghanistan Asia      13079460
6 Afghanistan Asia      14880372
```

range of columns

```r
# Select range of columns
sel_data <- gapminder %>% select(country, year:pop)
head(sel_data)
```

```
# A tibble: 6 x 4
  country      year lifeExp      pop
  <fct>       <int>   <dbl>    <int>
1 Afghanistan  1952    28.8  8425333
2 Afghanistan  1957    30.3  9240934
3 Afghanistan  1962    32.0 10267083
4 Afghanistan  1967    34.0 11537966
5 Afghanistan  1972    36.1 13079460
6 Afghanistan  1977    38.4 14880372
```

Exclude columns

```r
# Exclude columns
sel_data <- gapminder %>% select(-pop)
head(sel_data)
```

```
# A tibble: 6 x 5
  country     continent  year lifeExp gdpPercap
  <fct>       <fct>     <int>   <dbl>     <dbl>
1 Afghanistan Asia       1952    28.8      779.
2 Afghanistan Asia       1957    30.3      821.
3 Afghanistan Asia       1962    32.0      853.
4 Afghanistan Asia       1967    34.0      836.
5 Afghanistan Asia       1972    36.1      740.
6 Afghanistan Asia       1977    38.4      786.
```

Exclude multiple columns

```r
# Exclude multiple columns
sel_data <- gapminder %>% select(-pop, -country)
head(sel_data)
```

```
# A tibble: 6 x 4
  continent  year lifeExp gdpPercap
  <fct>     <int>   <dbl>     <dbl>
1 Asia       1952    28.8      779.
2 Asia       1957    30.3      821.
3 Asia       1962    32.0      853.
4 Asia       1967    34.0      836.
5 Asia       1972    36.1      740.
6 Asia       1977    38.4      786.
```

Exclude a vector of columns

```r
# Exclude a vector of columns
sel_data <- gapminder %>% select(-c(pop, country))
head(sel_data)
```

```
# A tibble: 6 x 4
  continent  year lifeExp gdpPercap
  <fct>     <int>   <dbl>     <dbl>
1 Asia       1952    28.8      779.
2 Asia       1957    30.3      821.
3 Asia       1962    32.0      853.
4 Asia       1967    34.0      836.
5 Asia       1972    36.1      740.
6 Asia       1977    38.4      786.
```

Exclude a Complement of columns

```r
# Complement of columns
f1 <- gapminder %>% select(!c(continent, gdpPercap))
head(f1)
```

```
# A tibble: 6 x 4
  country      year lifeExp      pop
  <fct>       <int>   <dbl>    <int>
1 Afghanistan  1952    28.8  8425333
2 Afghanistan  1957    30.3  9240934
3 Afghanistan  1962    32.0 10267083
```

```
4 Afghanistan  1967    34.0 11537966
5 Afghanistan  1972    36.1 13079460
6 Afghanistan  1977    38.4 14880372
```

Rename columns with select

```r
# Rename columns with select
sel_data <- gapminder %>% select(Life_Expectancy = lifeExp)
head(sel_data)
```

```
# A tibble: 6 x 1
  Life_Expectancy
            <dbl>
1            28.8
2            30.3
3            32.0
4            34.0
5            36.1
6            38.4
```

Rename columns with rename()

```r
# Rename columns with rename()
sel_data <- gapminder %>% rename(Life_Expectancy = lifeExp)
head(sel_data)
```

```
# A tibble: 6 x 6
  country     continent  year Life_Expectancy      pop gdpPercap
  <fct>       <fct>     <int>           <dbl>    <int>     <dbl>
1 Afghanistan Asia       1952            28.8  8425333      779.
2 Afghanistan Asia       1957            30.3  9240934      821.
3 Afghanistan Asia       1962            32.0 10267083      853.
4 Afghanistan Asia       1967            34.0 11537966      836.
5 Afghanistan Asia       1972            36.1 13079460      740.
6 Afghanistan Asia       1977            38.4 14880372      786.
```

Additional Selection Methods

Select columns using indices, ranges, or specific patterns.

Select by index

```r
# Select by index
gapminder %>% select(c(2, 3)) %>% head()
```

```
# A tibble: 6 x 2
  continent  year
  <fct>     <int>
1 Asia       1952
2 Asia       1957
3 Asia       1962
4 Asia       1967
5 Asia       1972
6 Asia       1977
```

Select by index range

```r
# Select by index range
gapminder %>% select(2:3) %>% head()
```

```
# A tibble: 6 x 2
  continent  year
  <fct>     <int>
1 Asia       1952
2 Asia       1957
3 Asia       1962
4 Asia       1967
5 Asia       1972
6 Asia       1977
```

Exclude by labels

```r
# Select by labels
gapminder %>% select(c('country', 'continent')) %>% head()
```

```
# A tibble: 6 x 2
  country     continent
  <fct>       <fct>
1 Afghanistan Asia
2 Afghanistan Asia
3 Afghanistan Asia
4 Afghanistan Asia
5 Afghanistan Asia
6 Afghanistan Asia
```

Exclude by labels

```r
# Exclude by labels
gapminder %>% select(-c('country', 'continent')) %>% head()
```

```
# A tibble: 6 x 4
   year lifeExp      pop gdpPercap
  <int>   <dbl>    <int>     <dbl>
1  1952    28.8  8425333      779.
2  1957    30.3  9240934      821.
3  1962    32.0 10267083      853.
4  1967    34.0 11537966      836.
5  1972    36.1 13079460      740.
6  1977    38.4 14880372      786.
```

Select columns ending with 'p'

```r
# Select columns starting with 'co'
gapminder %>% select(starts_with('co')) %>% head()
```

```
# A tibble: 6 x 2
  country     continent
  <fct>       <fct>
1 Afghanistan Asia
```

```
2 Afghanistan Asia
3 Afghanistan Asia
4 Afghanistan Asia
5 Afghanistan Asia
6 Afghanistan Asia
```

Select columns ending with 'p'

```r
# Select columns ending with 'p'
gapminder %>% select(ends_with('p')) %>% head()
```

```
# A tibble: 6 x 3
  lifeExp      pop gdpPercap
    <dbl>    <int>     <dbl>
1    28.8  8425333      779.
2    30.3  9240934      821.
3    32.0 10267083      853.
4    34.0 11537966      836.
5    36.1 13079460      740.
6    38.4 14880372      786.
```

Select columns containing 'pop'

```r
# Select columns containing 'pop'
gapminder %>% select(contains('pop')) %>% head()
```

```
# A tibble: 6 x 1
       pop
     <int>
1  8425333
2  9240934
3 10267083
4 11537966
5 13079460
6 14880372
```

Filtering Rows

Use `filter()` to subset rows based on conditions.

Filter for Belgium

```
# Filter for Belgium
filt_data <- gapminder %>% filter(country == "Belgium") %>% head(8)
filt_data
```

```
# A tibble: 8 x 6
  country continent  year lifeExp     pop gdpPercap
  <fct>   <fct>     <int>   <dbl>   <int>     <dbl>
1 Belgium Europe     1952    68   8730405     8343.
2 Belgium Europe     1957    69.2 8989111     9715.
3 Belgium Europe     1962    70.2 9218400    10991.
4 Belgium Europe     1967    70.9 9556500    13149.
5 Belgium Europe     1972    71.4 9709100    16672.
6 Belgium Europe     1977    72.8 9821800    19118.
7 Belgium Europe     1982    73.9 9856303    20980.
8 Belgium Europe     1987    75.4 9870200    22526.
```

Filter with multiple conditions

```
# Filter with multiple conditions
gapminder %>%
  filter(country == "Oman" & year > 1980 & year <= 2000) %>%
  head(8)
```

```
# A tibble: 4 x 6
  country continent  year lifeExp     pop gdpPercap
  <fct>   <fct>     <int>   <dbl>   <int>     <dbl>
1 Oman    Asia       1982    62.7 1301048    12955.
2 Oman    Asia       1987    67.7 1593882    18115.
3 Oman    Asia       1992    71.2 1915208    18617.
4 Oman    Asia       1997    72.5 2283635    19702.
```

Filter using %in% operator

```
# Filter using %in% operator
scandinavia <- c("Denmark", "Iceland", "Norway", "Sweden", "Finland")
res <- gapminder %>% filter(country %in% scandinavia)
head(res)
```

```
# A tibble: 6 x 6
  country continent  year lifeExp     pop gdpPercap
  <fct>   <fct>     <int>   <dbl>   <int>     <dbl>
1 Denmark Europe     1952    70.8 4334000     9692.
2 Denmark Europe     1957    71.8 4487831    11100.
3 Denmark Europe     1962    72.4 4646899    13583.
4 Denmark Europe     1967    73.0 4838800    15937.
5 Denmark Europe     1972    73.5 4991596    18866.
6 Denmark Europe     1977    74.7 5088419    20423.
```

Combining Select and Filter

Combine `select()` and `filter()` to extract specific data.

```
uk_gdp <- gapminder %>%
  filter(country == "United Kingdom") %>%
  select(country, year, gdpPercap)
head(uk_gdp)
```

```
# A tibble: 6 x 3
  country         year gdpPercap
  <fct>          <int>     <dbl>
1 United Kingdom  1952     9980.
2 United Kingdom  1957    11283.
3 United Kingdom  1962    12477.
4 United Kingdom  1967    14143.
5 United Kingdom  1972    15895.
6 United Kingdom  1977    17429.
```

Mutating Data

Use `mutate()` to create new columns and `transmute()` to keep only new columns.

Using Mutate to add new columns

```r
# Add new columns
gapminder %>%
  mutate(pop_million = pop / 1000000,
         countName = substring(country, 1, 3)) %>%
  head()
```

```
# A tibble: 6 x 8
  country     continent  year lifeExp      pop gdpPercap pop_million countName
  <fct>       <fct>     <int>   <dbl>    <int>     <dbl>       <dbl> <chr>
1 Afghanistan Asia       1952    28.8  8425333      779.        8.43 Afg
2 Afghanistan Asia       1957    30.3  9240934      821.        9.24 Afg
3 Afghanistan Asia       1962    32.0 10267083      853.       10.3  Afg
4 Afghanistan Asia       1967    34.0 11537966      836.       11.5  Afg
5 Afghanistan Asia       1972    36.1 13079460      740.       13.1  Afg
6 Afghanistan Asia       1977    38.4 14880372      786.       14.9  Afg
```

Using Mutate to Keep only new columns

```r
# Keep only new columns
gapminder %>% transmute(total_years = lifeExp * pop) %>% head()
```

```
# A tibble: 6 x 1
  total_years
        <dbl>
1  242658016.
2  280296010.
3  328515855.
4  392521603.
5  472011552.
6  571971739.
```

Using Mutate to Classify GDP per capita

```r
# Classify GDP per capita
gapminder %>%
  mutate(GDP_Category = ifelse(gdpPercap > 10000, "High", "Low")) %>%
  select(country, year, gdpPercap, GDP_Category) %>%
  head(10)
```

```
# A tibble: 10 x 4
   country      year gdpPercap GDP_Category
   <fct>       <int>     <dbl> <chr>
 1 Afghanistan  1952      779. Low
 2 Afghanistan  1957      821. Low
 3 Afghanistan  1962      853. Low
 4 Afghanistan  1967      836. Low
 5 Afghanistan  1972      740. Low
 6 Afghanistan  1977      786. Low
 7 Afghanistan  1982      978. Low
 8 Afghanistan  1987      852. Low
 9 Afghanistan  1992      649. Low
10 Afghanistan  1997      635. Low
```

Arranging Data

Sort data using `arrange()`.

Sort in ascending order

```r
# Sort in ascending order
gapminder %>% arrange(year, gdpPercap) %>% head()
```

```
# A tibble: 6 x 6
  country       continent  year lifeExp      pop gdpPercap
  <fct>         <fct>     <int>   <dbl>    <int>     <dbl>
1 Lesotho       Africa     1952    42.1   748747      299.
2 Guinea-Bissau Africa     1952    32.5   580653      300.
3 Eritrea       Africa     1952    35.9  1438760      329.
4 Myanmar       Asia       1952    36.3 20092996      331
5 Burundi       Africa     1952    39.0  2445618      339.
6 Ethiopia      Africa     1952    34.1 20860941      362.
```

Sort in descending order

```
# Sort in descending order
gapminder %>% arrange(year, desc(gdpPercap)) %>% head()
```

```
# A tibble: 6 x 6
  country       continent  year lifeExp       pop gdpPercap
  <fct>         <fct>     <int>   <dbl>     <int>     <dbl>
1 Kuwait        Asia       1952    55.6    160000   108382.
2 Switzerland   Europe     1952    69.6   4815000    14734.
3 United States Americas   1952    68.4 157553000    13990.
4 Canada        Americas   1952    68.8  14785584    11367.
5 New Zealand   Oceania    1952    69.4   1994794    10557.
6 Norway        Europe     1952    72.7   3327728    10095.
```

Summarizing Data

Compute summary statistics with `summarise()`.

```
scandinavia <- c("Denmark", "Iceland", "Norway", "Sweden", "Finland")
gapminder %>%
  filter(year == 1982 & country %in% scandinavia) %>%
  summarise(n_obs = n(),
            total_pop = sum(pop),
            mean_life_exp = mean(lifeExp),
            range_life_exp = max(lifeExp) - min(lifeExp))
```

```
# A tibble: 1 x 4
  n_obs total_pop mean_life_exp range_life_exp
  <int>     <int>         <dbl>          <dbl>
1     5  22618787          75.7           2.44
```

Grouping Data

Group data with `group_by()` for per-group summaries.

```r
gapminder %>%
  group_by(year) %>%
  summarise(n_obs = n(),
            n_countries = n_distinct(country),
            total_pop = sum(pop),
            mean_life_exp = mean(lifeExp),
            range_life_exp = max(lifeExp) - min(lifeExp))
```

```
# A tibble: 12 x 6
    year n_obs n_countries  total_pop mean_life_exp range_life_exp
   <int> <int>       <int>      <dbl>         <dbl>          <dbl>
 1  1952   142         142 2406957150          49.1           43.9
 2  1957   142         142 2664404580          51.5           43.1
 3  1962   142         142 2899782974          53.6           41.7
 4  1967   142         142 3217478384          55.7           40.1
 5  1972   142         142 3576977158          57.6           39.3
 6  1977   142         142 3930045807          59.6           44.9
 7  1982   142         142 4289436840          61.5           38.7
 8  1987   142         142 4691477418          63.2           38.8
 9  1992   142         142 5110710260          64.2           55.8
10  1997   142         142 5515204472          65.0           44.6
11  2002   142         142 5886977579          65.7           42.8
12  2007   142         142 6251013179          67.0           43.0
```

Joining Data

Perform joins using `inner_join()`, `left_join()`, `right_join()`, and `full_join()`.

Example data for joining

```r
# Create sample datasets
employees <- data.frame(
  id = c(1, 2, 3, 4),
  name = c("Alice", "Bob", "Charlie", "David"),
  age = c(25, 30, 22, 28)
)
```

```r
# Create sample datasets
departments <- data.frame(
  id = c(2, 3, 5, 6),
  department = c("HR", "Finance", "IT", "Sales")
)
```

Inner join

```r
# Perform inner join
employees %>% inner_join(departments, by = "id")
```

```
  id    name age department
1  2     Bob  30         HR
2  3 Charlie  22    Finance
```

left join

```r
# Perform left join
employees %>% left_join(departments, by = "id")
```

```
  id    name age department
1  1   Alice  25       <NA>
2  2     Bob  30         HR
3  3 Charlie  22    Finance
4  4   David  28       <NA>
```

right join

```r
# Perform right join
employees %>% right_join(departments, by = "id")
```

```
  id    name age department
1  2     Bob  30         HR
2  3 Charlie  22    Finance
3  5    <NA>  NA         IT
4  6    <NA>  NA      Sales
```

full join

```
# Perform full join
employees %>% full_join(departments, by = "id")
```

```
  id    name age department
1  1   Alice  25       <NA>
2  2     Bob  30         HR
3  3 Charlie  22    Finance
4  4   David  28       <NA>
5  5    <NA>  NA         IT
6  6    <NA>  NA      Sales
```

Reading and Writing Files

Use `readr` to read and write delimited files, and `openxlsx` for Excel files.

Write to a file

```
# Write to a file
write_delim(res, "./methods_for_data_scince/applied_R/episodes/data/gapminder2.txt", delim =
```

Write to Excel

```
# Write to Excel
write.xlsx(res, file = "./methods_for_data_scince/applied_R/episodes/data//gapm3.xlsx", shee
```

Read from Excel

```
# Read from Excel
xl <- read.xlsx("./methods_for_data_scince/applied_R/episodes/data//gapm3.xlsx", sheet = "ga
```

Tidying Data with tidyr

Handle missing values using `tidyr`.

Sample data with missing values

```
# Sample data with missing values
df <- data.frame(
  Name = c("Alice", "Bob", "Charlie", NA, "Eve"),
  Age = c(25, NA, 30, 28, 22),
  Score = c(80, 90, NA, 75, 95)
)
```

Remove rows with any NA

```
# Remove rows with any NA
df_clean <- drop_na(df)
```

Remove rows with NA in specific column

```
# Remove rows with NA in specific column
df_clean_age <- drop_na(df, Age)
```

Replace NA values

```
# Replace NA values
df_filled <- df %>% replace_na(list(Age = 0, Score = 50, Name = "Unknown"))
```

String Manipulation with stringr

Perform string operations using `stringr`.

Split string with stringr

```
# Split string
text <- "Hello world, welcome to R programming!"
words <- str_split(text, " ")
print(words)
```

```
[[1]]
[1] "Hello"        "world,"       "welcome"      "to"           "R"
[6] "programming!"
```

Here we split the sentence up into a list of words

Access specific word with stringr

```
# Access specific word
print(words[[1]][3])
```

```
[1] "welcome"
```

We can print specific works in the list

Extract substring with stringr

bedwqbblsmuv#| echo: true #| echo: true # Extract substring sub_text <- str_sub(text, 7, 11) print(sub_text)

we can extract a substring

Replace substring with stringr

```
# Replace substring
str_sub(text, 1, 5) <- "Hi"
print(text)
```

```
[1] "Hi world, welcome to R programming!"
```

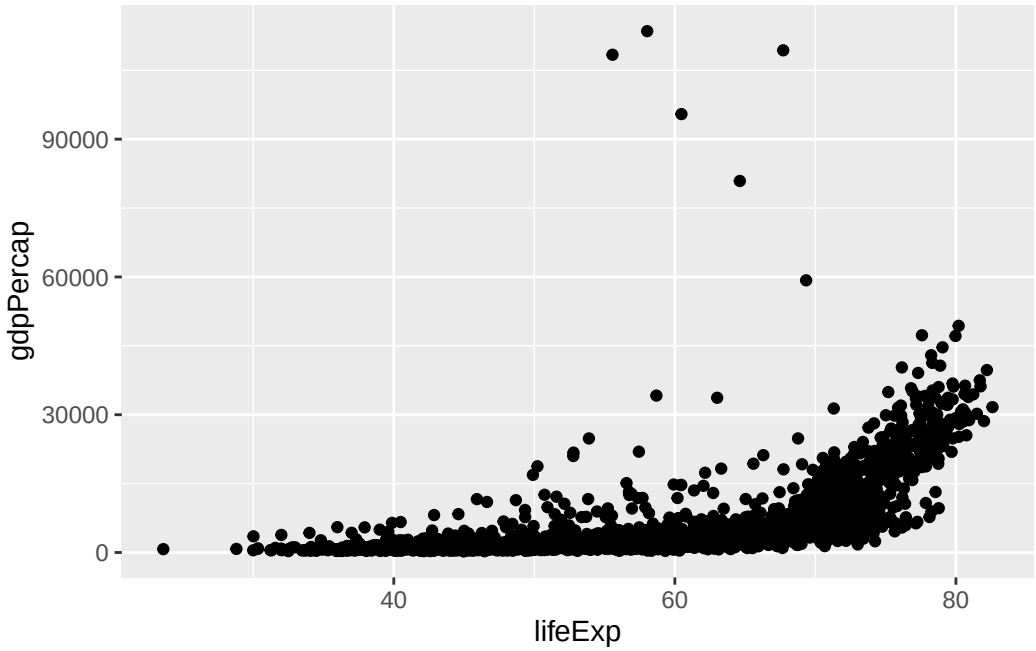
and we can replace a substring directly

Visualizations with ggplot2

Create visualizations using `ggplot2`.

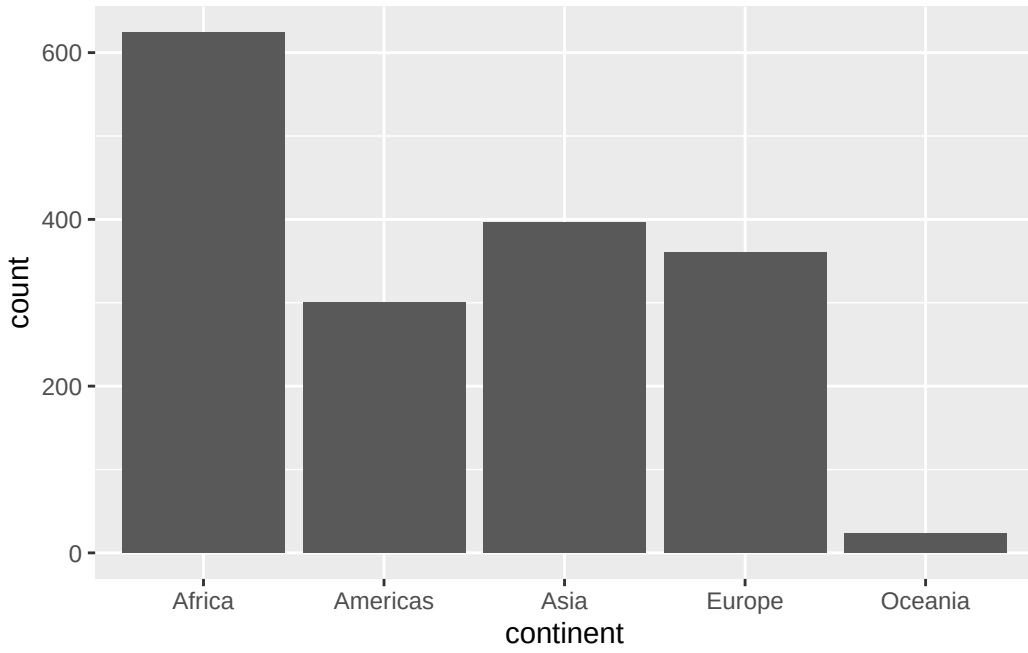
Scatter plot with ggplot2

```
# Basic scatter plot
ggplot(data = gapminder, aes(x = lifeExp, y = gdpPercap)) +
  geom_point()
```



Bar plot with ggplot2

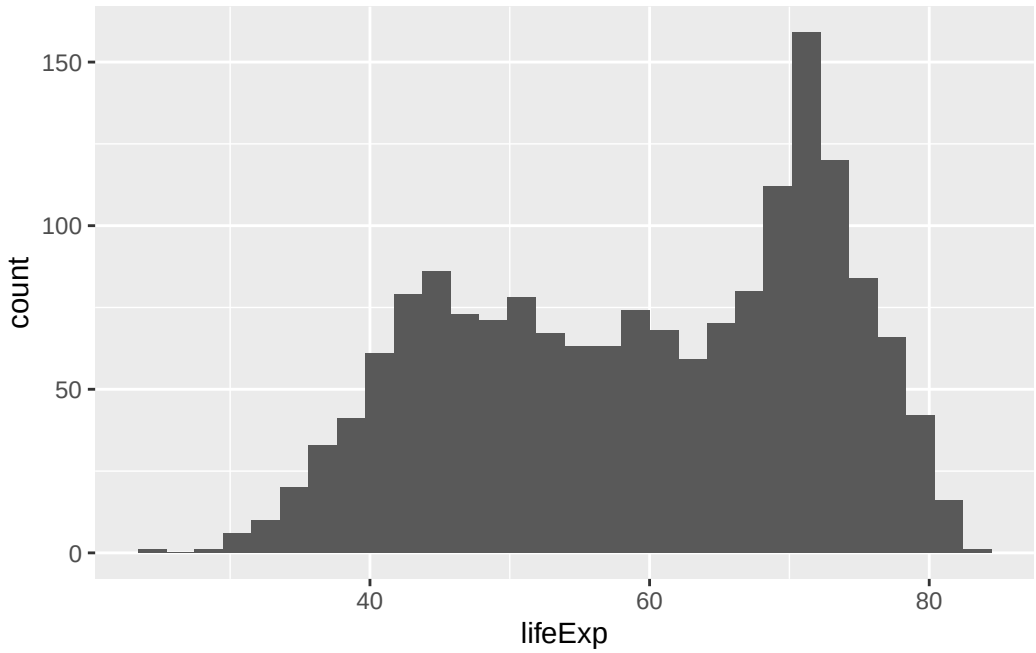
```
# Bar plot
ggplot(data = gapminder, aes(x = continent)) +
  geom_bar()
```



Histogram plot with ggplot2

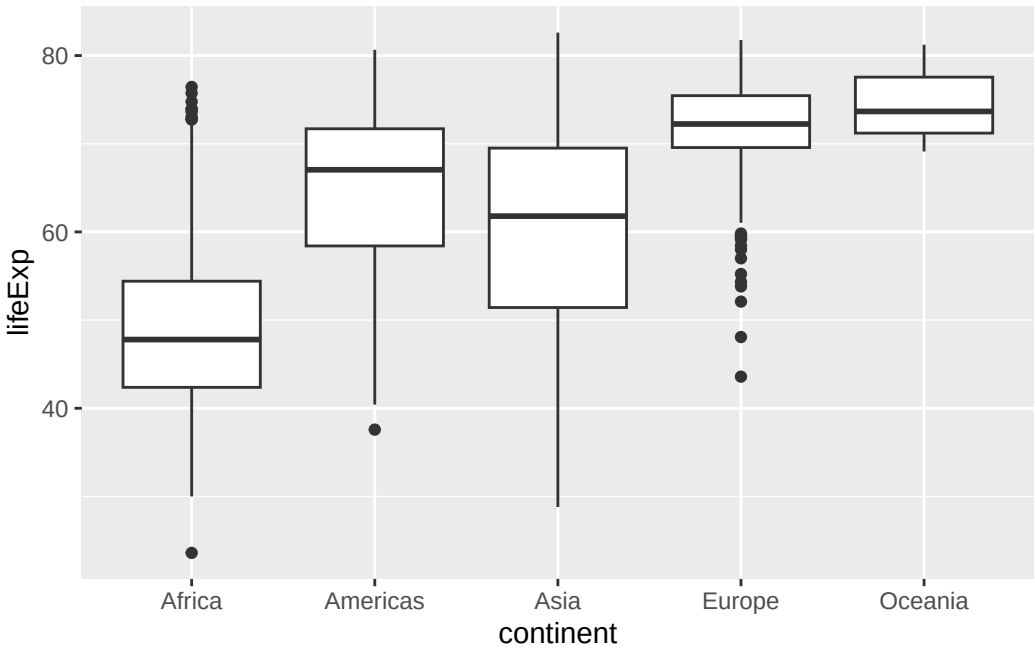
```
# Histogram
ggplot(data = gapminder, aes(x = lifeExp)) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



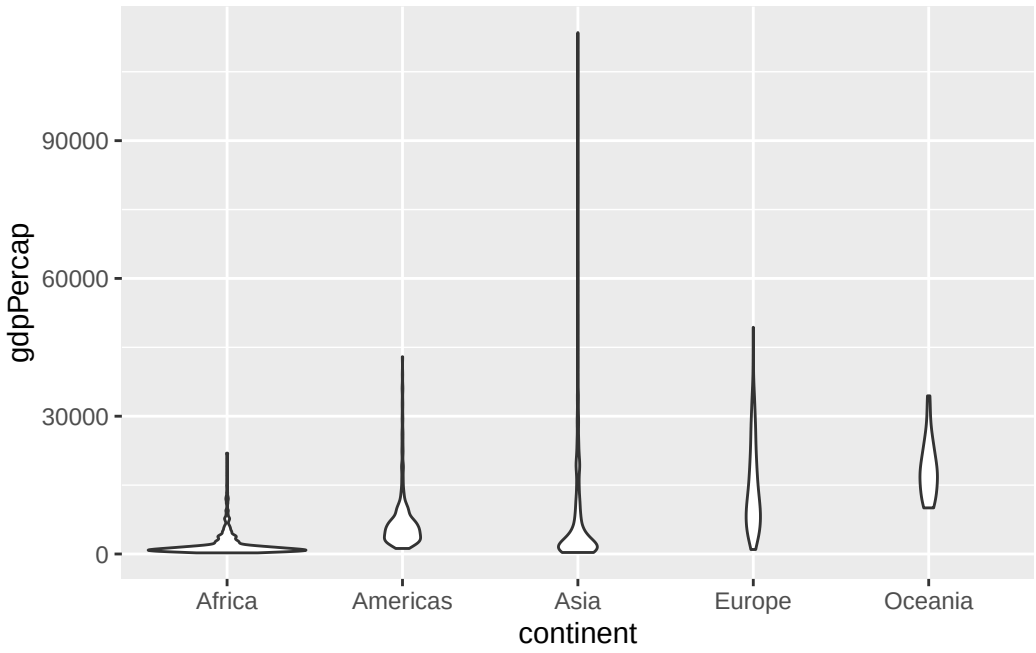
Box plot with ggplot2

```
# Box plot
ggplot(gapminder, aes(x = continent, y = lifeExp)) +
  geom_boxplot()
```



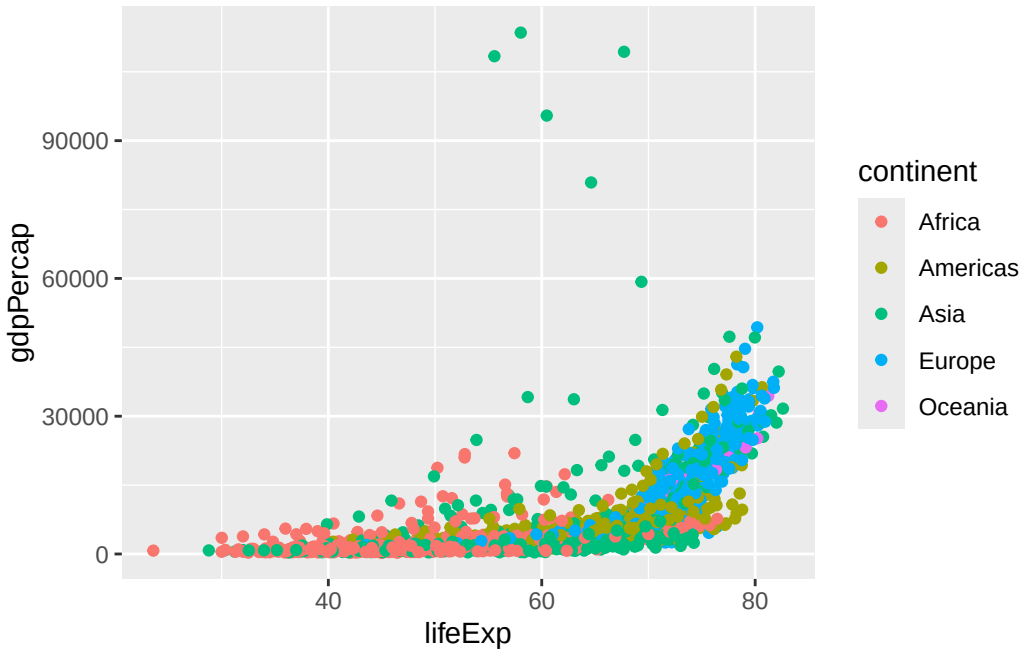
Violin plot with ggplot2

```
# Violin plot
ggplot(gapminder, aes(x = continent, y = gdpPercap)) +
  geom_violin()
```



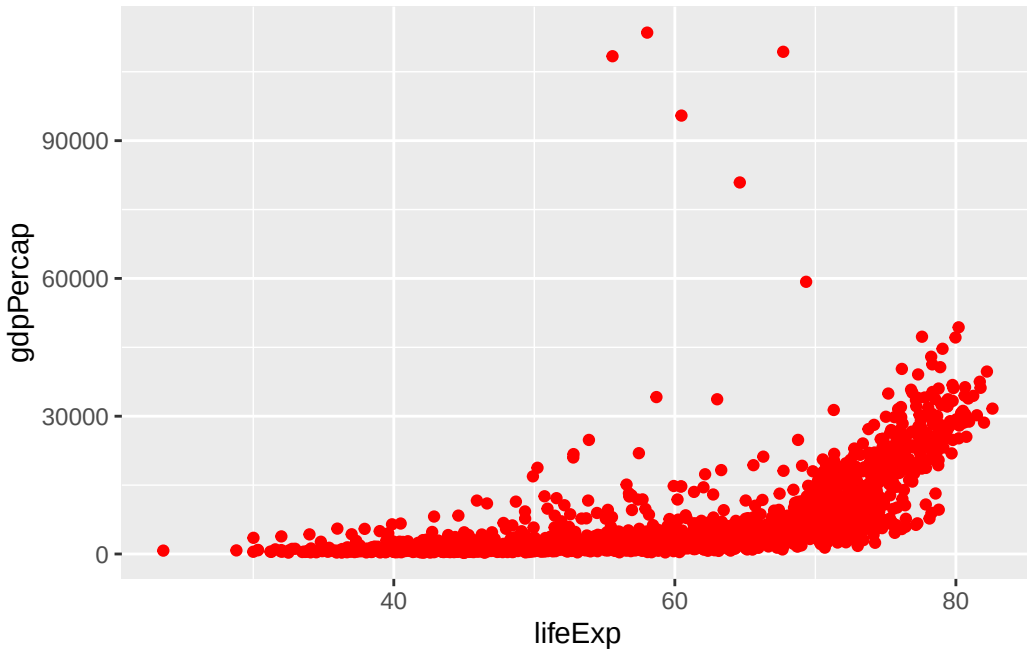
Scatter plot with color aesthetic with ggplot2

```
# Scatter plot with color aesthetic
ggplot(data = gapminder, aes(x = lifeExp, y = gdpPercap, color = continent)) +
  geom_point()
```



Scatter plot with custom color

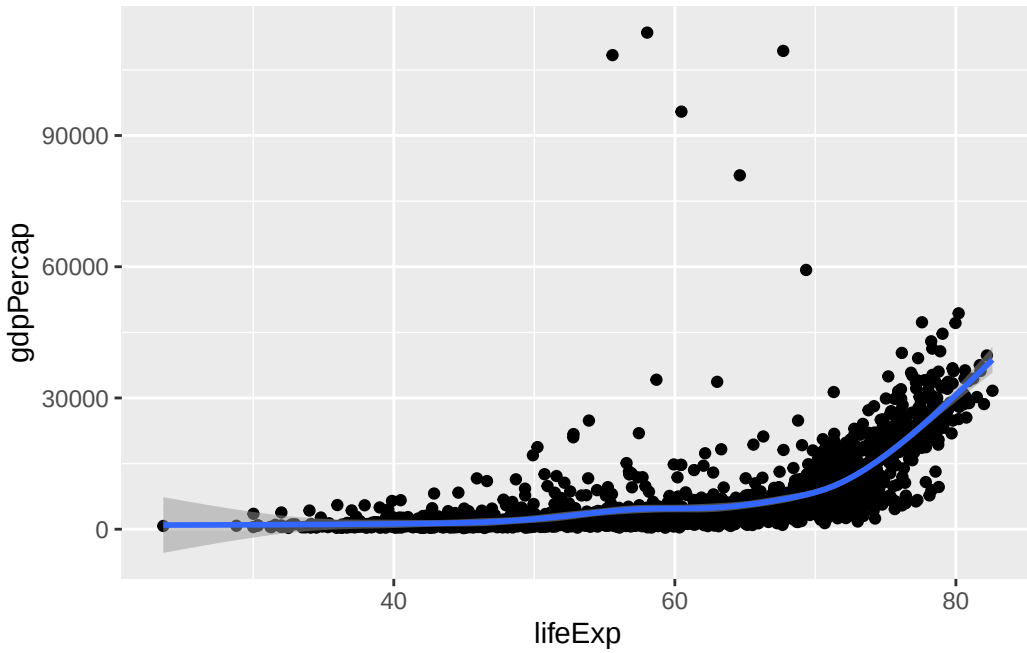
```
# Scatter plot with custom color
ggplot(data = gapminder, aes(x = lifeExp, y = gdpPercap)) +
  geom_point(color = "red")
```



Scatter plot with smooth line

```
# Scatter plot with smooth line
ggplot(data = gapminder, aes(x = lifeExp, y = gdpPercap)) +
  geom_point() +
  geom_smooth(alpha = 0.5)
```

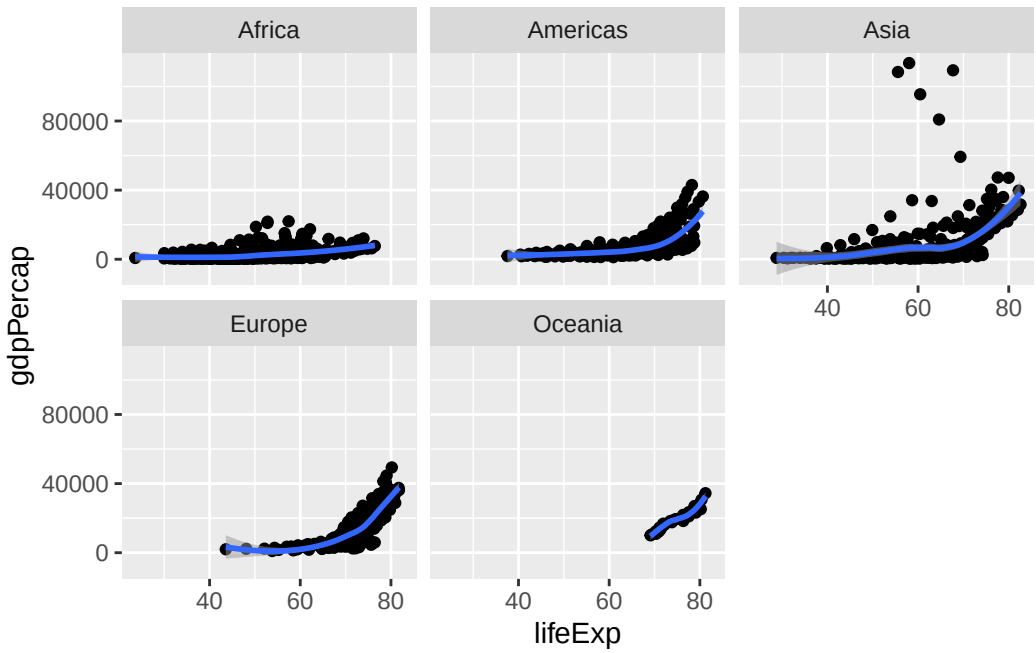
```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```



Faceted plot in ggplot2

```
# Faceted plot
ggplot(data = gapminder, aes(x = lifeExp, y = gdpPercap)) +
  geom_point() +
  geom_smooth(alpha = 0.5) +
  facet_wrap(~continent)
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



Customized axes

```
# Customized axes
ggplot(data = gapminder, aes(x = lifeExp, y = gdpPercap)) +
  geom_point() +
  geom_smooth(alpha = 0.5) +
  xlab("Life Expectancy") +
  ylab("GDP Per Capita") +
  xlim(0, 70) +
  ylim(0, 100000)
```

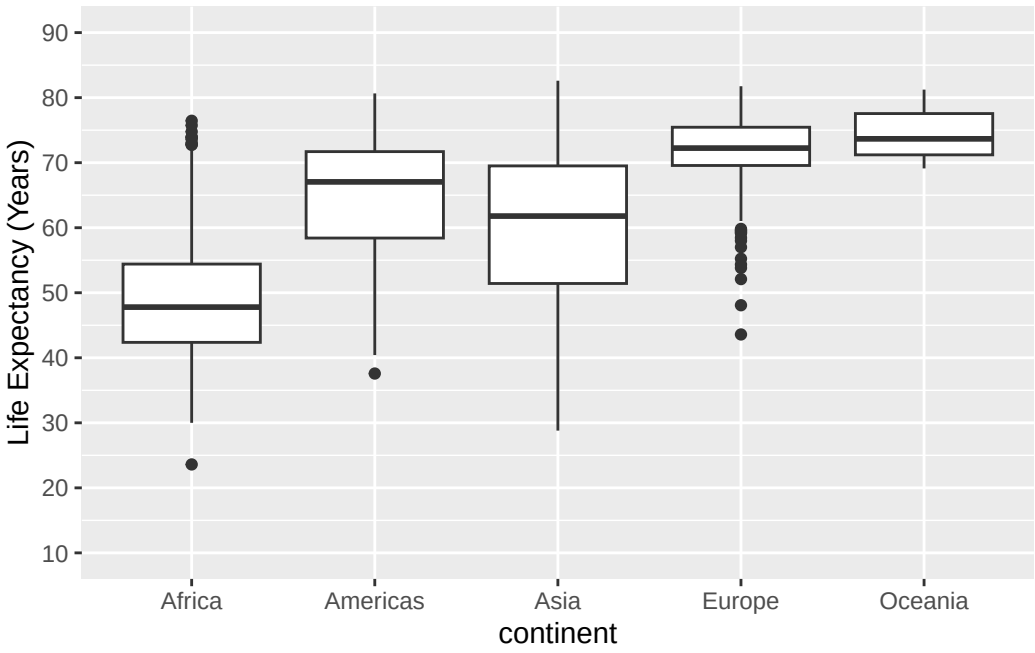
```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'

Warning: Removed 496 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 496 rows containing missing values or values outside the scale range
(`geom_point()`).
```

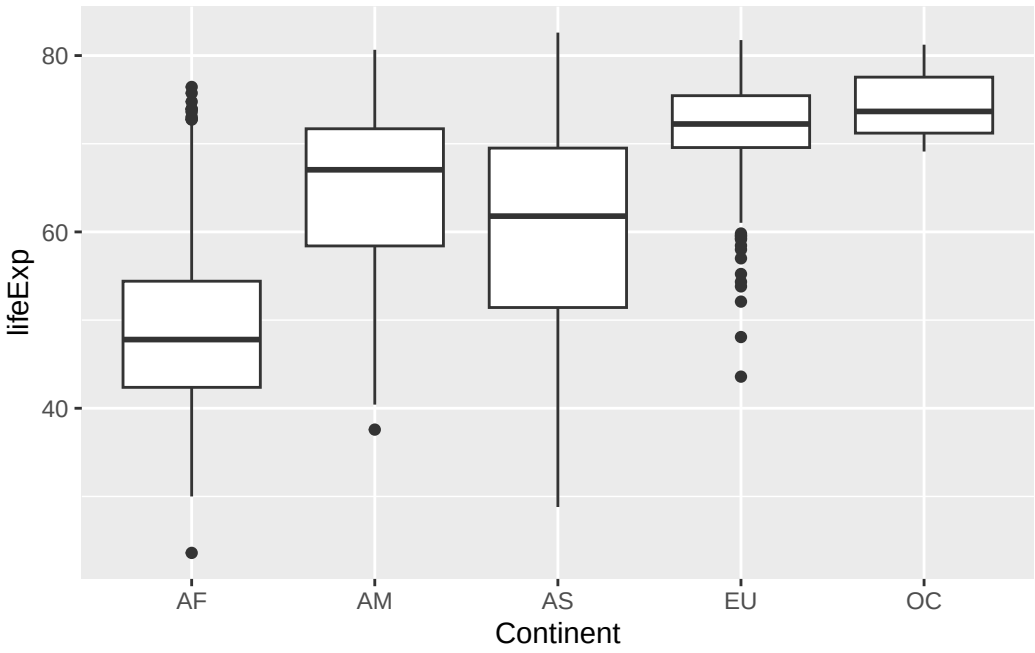
Box plot with customized y-axis

```
# Box plot with customized y-axis
ggplot(gapminder, aes(x = continent, y = lifeExp)) +
  geom_boxplot() +
  scale_y_continuous(name = "Life Expectancy (Years)",
                     limits = c(10, 90),
                     breaks = c(10, 20, 30, 40, 50, 60, 70, 80, 90))
```



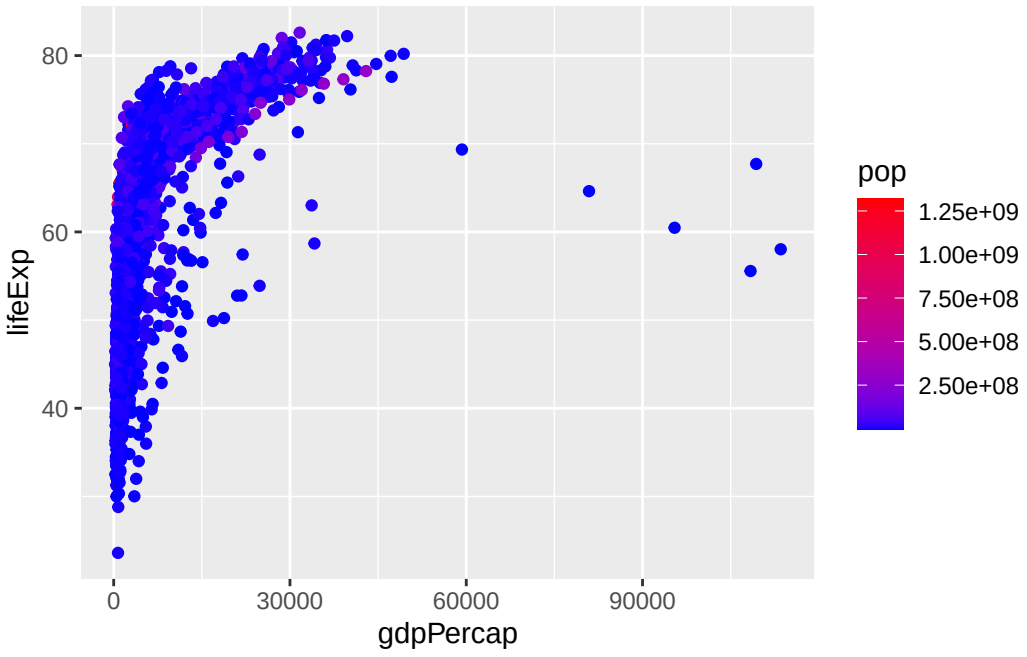
Box plot with customized x-axis

```
# Box plot with customized x-axis
ggplot(gapminder, aes(x = continent, y = lifeExp)) +
  geom_boxplot() +
  scale_x_discrete(name = "Continent",
                   labels = c("Africa" = "AF", "Americas" = "AM",
                              "Asia" = "AS", "Europe" = "EU", "Oceania" = "OC"))
```



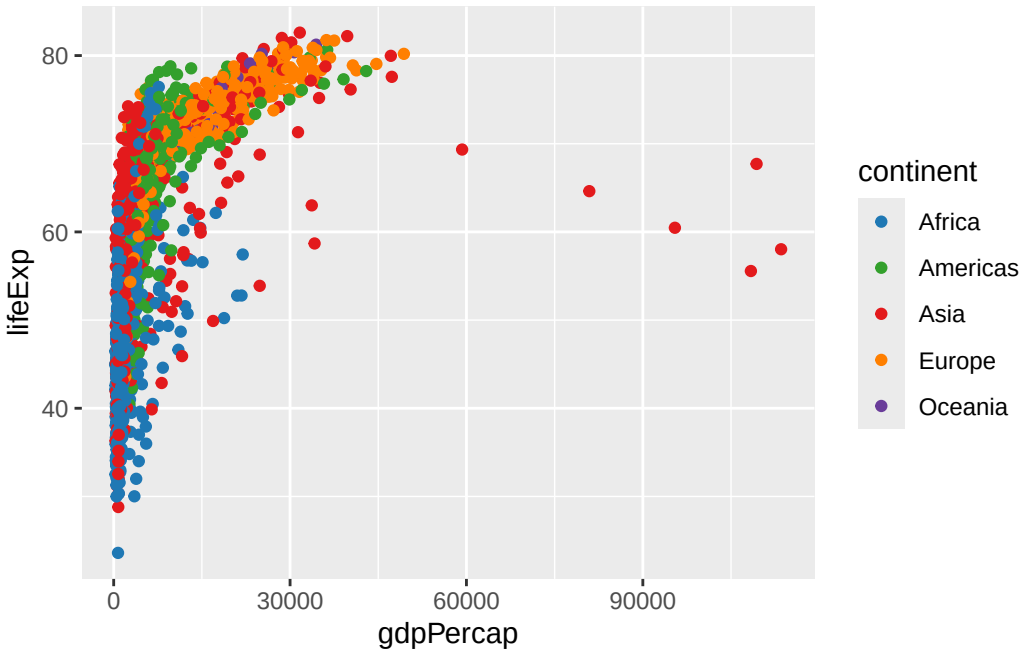
Color gradient scale

```
# Color gradient scale
ggplot(data = gapminder, aes(x = gdpPercap, y = lifeExp, color = pop)) +
  geom_point() +
  scale_color_gradient(low = "blue", high = "red")
```



Discrete color scale

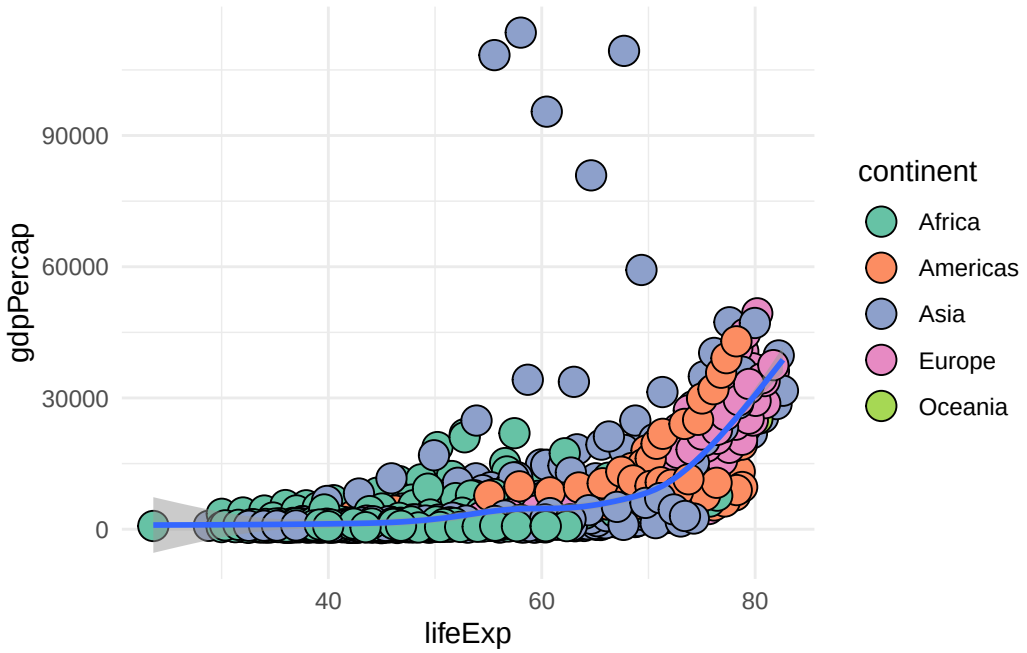
```
# Discrete color scale
ggplot(data = gapminder, aes(x = gdpPercap, y = lifeExp, color = continent)) +
  geom_point() +
  scale_color_manual(values = c("#1f78b4", "#33a02c", "#e31a1c", "#ff7f00", "#6a3d9a"))
```



Plot with color palette

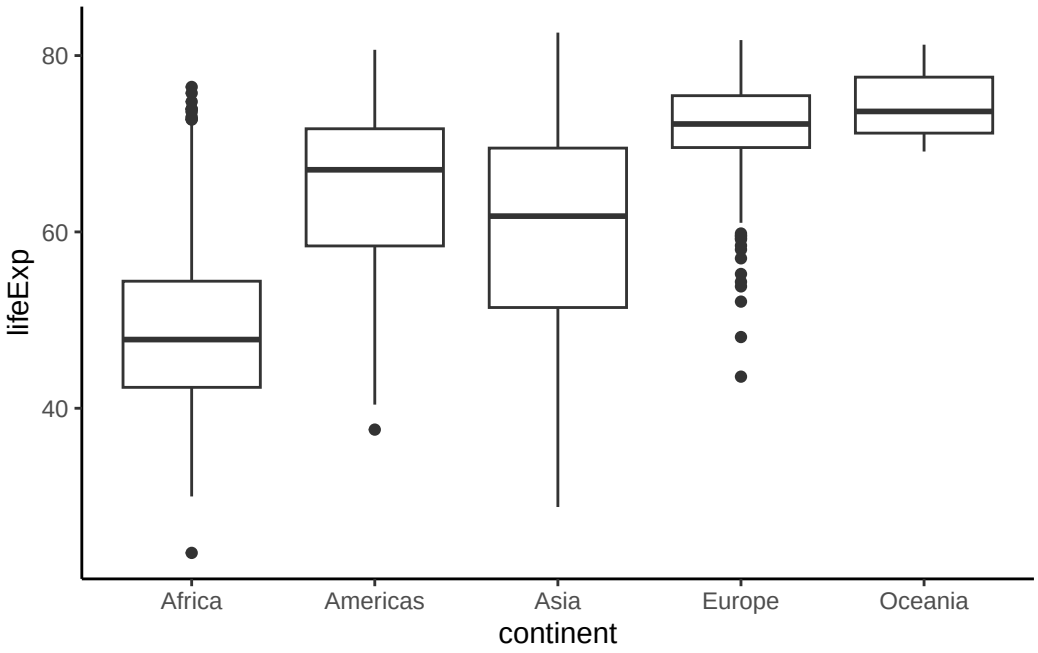
```
# Plot with color palette
ggplot(data = gapminder, aes(x = lifeExp, y = gdpPercap)) +
  geom_point(aes(fill = continent), size = 5, shape = 21) +
  geom_smooth(alpha = 0.5) +
  scale_fill_brewer(palette = "Set2") +
  theme_minimal()
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```



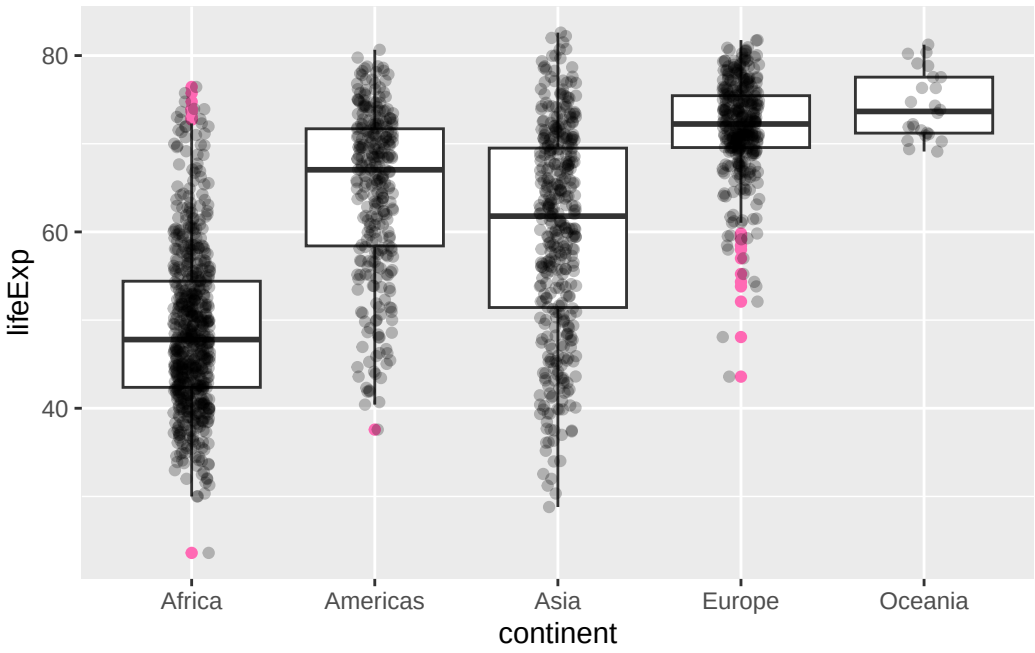
Box plot with theme_classic

```
# Box plot with theme_classic
ggplot(gapminder, aes(x = continent, y = lifeExp)) +
  geom_boxplot() +
  theme_classic()
```



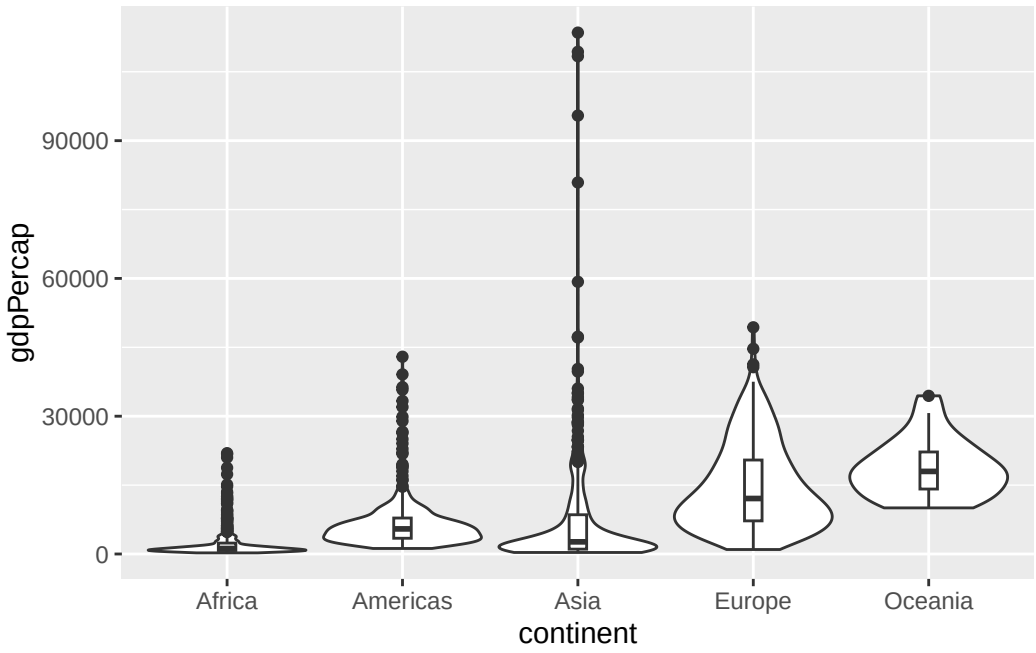
Box plot with jitter

```
# Box plot with jitter
ggplot(gapminder, aes(x = continent, y = lifeExp)) +
  geom_boxplot(outlier.colour = "hotpink") +
  geom_jitter(position = position_jitter(width = 0.1, height = 0), alpha = 1/4)
```



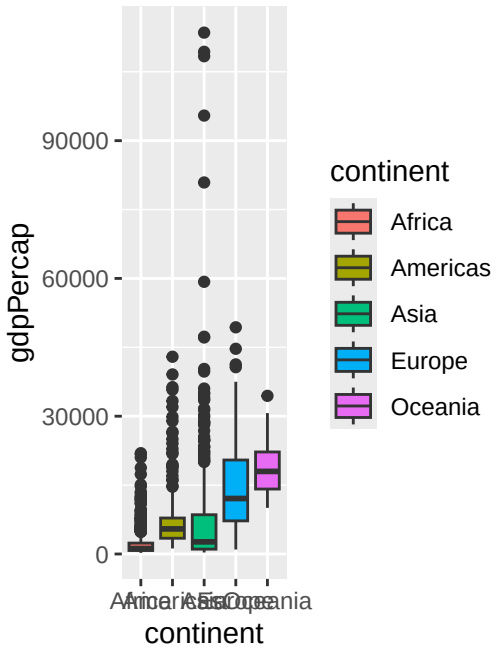
Combined violin and box plot

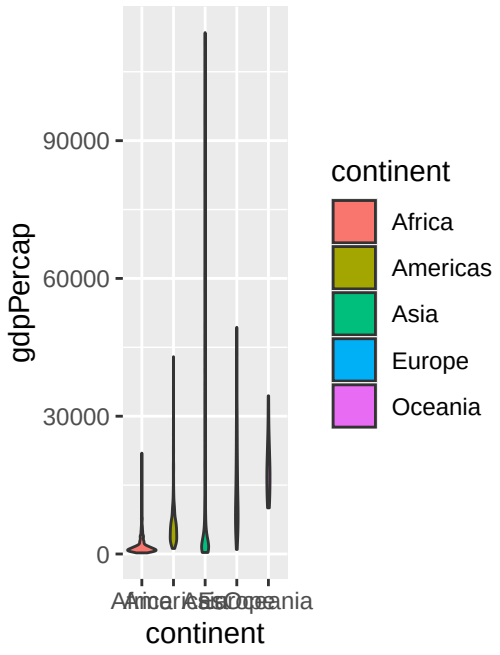
```
# Combined violin and box plot
ggplot(gapminder) +
  geom_violin(aes(x = continent, y = gdpPercap), scale = "width") +
  geom_boxplot(aes(x = continent, y = gdpPercap), width = 0.1)
```



Multiple plots with gridExtra

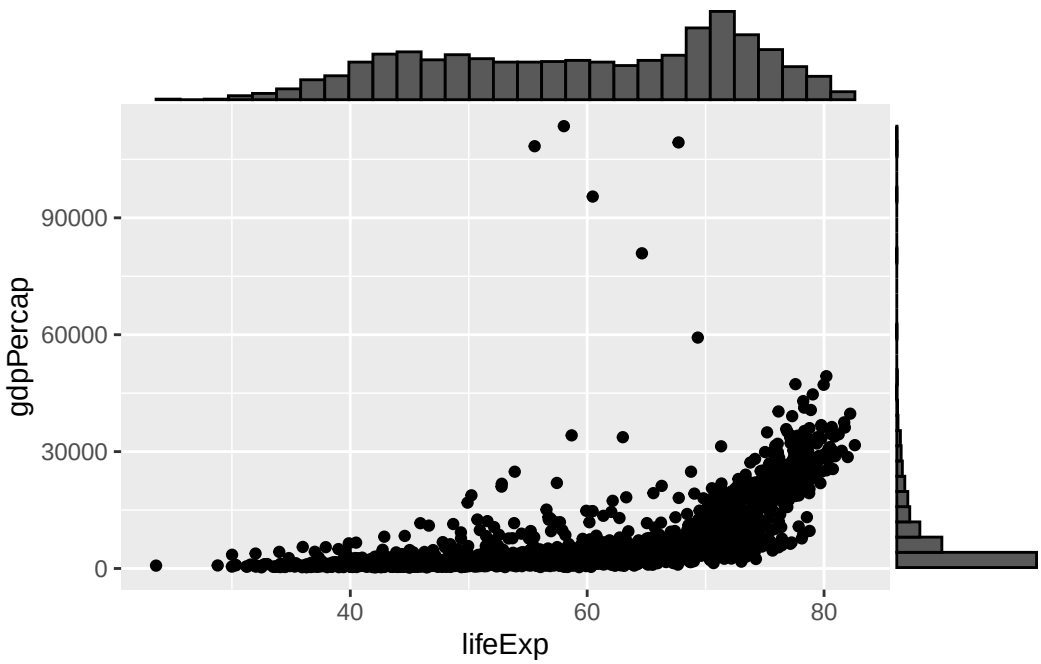
```
# Multiple plots with gridExtra
p1 <- ggplot(gapminder) + geom_boxplot(aes(x = continent, y = gdpPercap, fill = continent))
p2 <- ggplot(gapminder) + geom_violin(aes(x = continent, y = gdpPercap, fill = continent))
grid.arrange(p1, p2, nrow = 1)
```





Marginal histogram

```
# Marginal histogram
p <- ggplot(data = gapminder, aes(x = lifeExp, y = gdpPercap, fill = continent)) +
  geom_point() +
  theme(legend.position = "none")
ggMarginal(p, type = "histogram")
```



Saving Plots

Save a plot to a file using `ggsave`.

```
p1 <- ggplot(data = gapminder, aes(x = lifeExp, y = gdpPercap)) +
  geom_point(aes(fill = continent), size = 5, shape = 21) +
  geom_smooth(alpha = 0.5) +
  scale_fill_brewer(palette = "Set2") +
  theme_minimal()

ggsave("myplot.png", plot = p1)
```

```
Saving 5.5 x 3.5 in image
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

Exercise Solutions

Below are the solutions to the provided exercises.

Exercise 1: Select year, lifeExp, and country

```
gapminder %>% select(year, lifeExp, country) %>% head()
```

```
# A tibble: 6 x 3
   year lifeExp country
  <int>   <dbl> <fct>
1  1952    28.8 Afghanistan
2  1957    30.3 Afghanistan
3  1962    32.0 Afghanistan
4  1967    34.0 Afghanistan
5  1972    36.1 Afghanistan
6  1977    38.4 Afghanistan
```

Exercise 2: Select all columns except year, lifeExp, and country

```
gapminder %>% select(-year, -lifeExp, -country) %>% head()
```

```
# A tibble: 6 x 3
  continent      pop gdpPercap
  <fct>        <int>     <dbl>
1 Asia       8425333      779.
2 Asia       9240934      821.
3 Asia      10267083      853.
4 Asia      11537966      836.
5 Asia      13079460      740.
6 Asia      14880372      786.
```

Exercise 3: Filter for year 2002

```
gapminder %>% filter(year == 2002) %>% head()
```

```
# A tibble: 6 x 6
  country     continent  year lifeExp      pop gdpPercap
  <fct>       <fct>     <int>   <dbl>    <int>     <dbl>
1 Afghanistan Asia       2002    42.1 25268405      727.
```

```
2 Albania     Europe     2002    75.7  3508512     4604.
3 Algeria     Africa     2002    71.0 31287142     5288.
4 Angola      Africa     2002    41.0 10866106     2773.
5 Argentina   Americas   2002    74.3 38331121     8798.
6 Australia   Oceania    2002    80.4 19546792    30688.
```

Exercise 4: Store and display data for Asian countries

```r
gapminder_asia <- gapminder %>% filter(continent == "Asia")
head(gapminder_asia)
```

```
# A tibble: 6 x 6
  country     continent  year lifeExp      pop gdpPercap
  <fct>       <fct>     <int>   <dbl>    <int>     <dbl>
1 Afghanistan Asia       1952    28.8  8425333      779.
2 Afghanistan Asia       1957    30.3  9240934      821.
3 Afghanistan Asia       1962    32.0 10267083      853.
4 Afghanistan Asia       1967    34.0 11537966      836.
5 Afghanistan Asia       1972    36.1 13079460      740.
6 Afghanistan Asia       1977    38.4 14880372      786.
```

Exercise 5: Compare Ireland and Brazil in 1977

```r
# Ireland
gapminder %>% filter(year == 1977, country == "Ireland")
```

```
# A tibble: 1 x 6
  country continent  year lifeExp     pop gdpPercap
  <fct>   <fct>     <int>   <dbl>   <int>     <dbl>
1 Ireland Europe     1977    72.0 3271900    11151.
```

```r
# Brazil
gapminder %>% filter(year == 1977, country == "Brazil")
```

```
# A tibble: 1 x 6
  country continent  year lifeExp       pop gdpPercap
  <fct>   <fct>     <int>   <dbl>     <int>     <dbl>
1 Brazil  Americas   1977    61.5 114313951     6660.
```

Exercise 6: African countries in 2012 with specific criteria

```
df <- gapminder %>%
  filter(continent == "Africa" & year == 2012 & pop <= 3 & lifeExp >= 70) %>%
  select(country, continent)
df
```

```
# A tibble: 0 x 2
# i 2 variables: country <fct>, continent <fct>
```

Exercise 7: Lowest life expectancy

```
gapminder %>% arrange(lifeExp) %>% head(1)
```

```
# A tibble: 1 x 6
  country continent  year lifeExp     pop gdpPercap
  <fct>   <fct>     <int>   <dbl>   <int>     <dbl>
1 Rwanda  Africa     1992    23.6 7290203      737.
```

Exercise 8: Highest life expectancy

```
gapminder %>% arrange(desc(lifeExp)) %>% head(1)
```

```
# A tibble: 1 x 6
  country continent  year lifeExp       pop gdpPercap
  <fct>   <fct>     <int>   <dbl>     <int>     <dbl>
1 Japan   Asia       2007    82.6 127467972    31656.
```

Exercise 9: Shortest life expectancy in the Americas in 1962

```
gapminder %>%
  mutate(lifeExpMonths = 12 * lifeExp) %>%
  filter(year == 1962, continent == "Americas") %>%
  arrange(lifeExpMonths) %>% head(1)
```

```
# A tibble: 1 x 7
  country continent  year lifeExp     pop gdpPercap lifeExpMonths
  <fct>   <fct>     <int>   <dbl>   <int>     <dbl>         <dbl>
1 Bolivia Americas   1962    43.4 3593918     2181.          521.
```

Exercise 10: Summary statistics for life expectancy

```r
gapminder %>%
  summarise(
    lifeExp_min = min(lifeExp),
    lifeExp_max = max(lifeExp),
    lifeExp_mean = mean(lifeExp)
  )
```

```
# A tibble: 1 x 3
  lifeExp_min lifeExp_max lifeExp_mean
        <dbl>       <dbl>        <dbl>
1        23.6        82.6         59.5
```

Exercise 11: Life expectancy by year

```r
gapminder %>%
  group_by(year) %>%
  summarise(
    lifeExp_min = min(lifeExp),
    lifeExp_max = max(lifeExp),
    lifeExp_mean = mean(lifeExp)
  )
```

```
# A tibble: 12 x 4
    year lifeExp_min lifeExp_max lifeExp_mean
   <int>       <dbl>       <dbl>        <dbl>
 1  1952        28.8        72.7         49.1
 2  1957        30.3        73.5         51.5
 3  1962        32.0        73.7         53.6
 4  1967        34.0        74.2         55.7
 5  1972        35.4        74.7         57.6
 6  1977        31.2        76.1         59.6
```

```
 7  1982        38.4        77.1         61.5
 8  1987        39.9        78.7         63.2
 9  1992        23.6        79.4         64.2
10  1997        36.1        80.7         65.0
11  2002        39.2        82           65.7
12  2007        39.6        82.6         67.0
```

Exercise 12: Explore join statements

Already demonstrated in the "Joining Data" section above.

Exercise 13: Read a file

This requires a file like `gapminder.xlsx`. Assuming it's available:

```
xl <- read.xlsx("./methods_for_data_scince/applied_R/episodes/data/gapm3.xlsx", sheet = "gap
head(xl)
```

Exercise 14: Get and split working directory path

```
# Get working directory
wd <- getwd()
# Split by appropriate delimiter
path_parts <- str_split(wd, "/")[[1]]
# Fetch last element
last_element <- path_parts[length(path_parts)]
print(last_element)
```

```
[1] "episodes"
```

Conclusion

This document covers key `tidyverse` functionalities and `ggplot2` visualizations using the `gapminder` dataset. It includes data manipulation, file handling, string manipulation, and various plotting techniques, along with solutions to the exercises.

To download this R Markdown file, save it with a `.Rmd` extension and open it in RStudio or another R Markdown-compatible environment.